

Ziad Ashraf

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AI Engineer with 2+ years of experience applying **Data Science**, **AI** and **ML** to solve real-world business problems. Hands-on experience in **predictive modeling**, **NLP**, **Speech AI**, **deep learning**, and end-to-end AI system development. Skilled in taking AI solutions from research and prototyping to production using modern **software engineering** principles, **MLOps**, and cloud-based deployment workflows.

Experience

Elshayeb

AI Engineer

Feb 2026 – Present

Alexandria | Remote

- Architected end-to-end **AI solutions** for **media operations** and **CRM** workflows, automating manual processes and significantly reducing task completion time.
- Engineered an intelligent lead qualification service using Gemini to generate **tenant-specific onboarding questionnaires**, enforcing **Pydantic** structured output schemas, **context-constrained** question generation, and **Langfuse** for **prompt versioning** and **token/cost optimization**.
- Designed a **two-stage multi-tenant SaaS lead retrieval system** leveraging **hybrid search (BM25 + dense embeddings)**, **reranking**, and a custom relevance scoring algorithm to match lead profile answers with tenant-specific questionnaires, improving retrieval accuracy and search quality over baseline hard filters.
- Architected and integrated an **Egyptian Arabic ASR** Adobe Premiere Pro plugin for **in-house video editors**, combining **VAD**, **CTC forced alignment**, and **Multimodal LLM** to achieve an **8%** macro WER (**3%** CER) and cut transcription costs by **~66%**.
- Implemented a **Gemini**-powered sales analytics and **TTS** reporting service that analyzes sales representative performance, identifies short-term trends, and generates automated voice reports, reducing sales managers' time spent reviewing large teams.

ISFP

Data Engineer / BI Developer

Jul 2026 – Present

Alexandria | Remote

- Delivered** end-to-end **operational data and analytics solutions** for maritime logistics and government port authorities across the **Suez Canal Economic Zone**.
- Engineered** automated **ETL pipelines** in **Alteryx** and **Oracle SQL** featuring **incremental loading** and automated profiling checks, optimizing complex queries and structuring **data marts** for high-integrity processing.
- Deployed** interactive **Tableau Server dashboards** integrated across multi-source operational data, applying statistical analysis to track critical port **KPIs**.
- Implemented** secure **Tableau Connected Apps** authentication via **JWT**, parsing custom payload claims to dynamically enforce **row-level security (RLS)** and restrict unauthorized navigation across port portals.

SimpliLearn Education Consulting Private Limited

Data Scientist

Dec 2022 – Jun 2024

Alexandria | Remote

- Delivered **data science and applied analytics solutions** for academic and research clients, completing 15+ capstone implementations with a **95% client acceptance rate**.
- Optimized ML models using **Python** and **R**, cutting baseline prediction error by **5–30%**, through hyperparameter tuning and feature engineering.
- Built **interactive dashboards** using **Power BI**, **Tableau** translating complex client data into dynamic visual metrics and exploratory analytics.
- Maintained a **100%** on-time delivery rate across all client milestones, providing ongoing project support that drove a **25%** increase in client retention.

Freelancer.com

Data Scientist

Jun 2022 – Apr 2022

Alexandria | Remote

- Deployed ML system on **AWS (EC2)** with automated retraining pipeline using **Python** and **Linux** achieving **100%** error-free deployment.
- Implemented 2 real-time IoT **ETL** pipelines and dashboards in **Zoho Analytics** to monitor the daily efficiency of **50+** devices, which reduced data errors by **90%**.
- Validated 3 research studies in **SPSS**, achieving over **90%** reliability, which supported evidence-based policy decisions.
- Received 5-star ratings from 8 clients for delivering high-quality, on-time results with comprehensive documentation and responsive support.

Projects

Real Estate Intelligence System

Oct 2025 – Sep 2026

Tech Stack: *Python, FastAPI, Docker, AWS, GitHub Actions, ONNX, Milvus, Prometheus, Grafana, Dagster, ClickHouse, Mlflow, RAGAS, Redis*

- Engineered** **Dagster** ETL and **scraping pipeline** on **Docker**, ingesting 10.9K+ Egyptian property listings into **ClickHouse** with automated **Pydantic** validation, achieving a **94.0%** data quality score.
- Architected a multi-stage **hybrid search** architecture integrating LLM structured filter extraction, INT8 multilingual-e5, Milvus BM25, and an ONNX bge-reranker, boosting **MRR** by +592% (0.059 → 0.412) and **NDCG@10** by +330% (0.052 → 0.223) over lexical baselines.
- Benchmarked generation quality using automated **RAGAS** evaluation suites and **Mlflow GenAI Tracing**, validating end-to-end response fidelity to record an **82.1% Answer Relevancy score**.
- Implemented a **FastAPI** serving layer with **JWT/RBAC**, sliding-window rate limiting, and a two-tier cache (Redis + Milvus semantic), eliminating redundant LLM compute to drop P50 latency from **11.6s** to **<10ms**.

ContentLens AI: AI-Powered Marketing Document Platform

Dec 2025 – Feb 2026

Tech Stack: *Python, FastAPI, LangChain, LangGraph, Langfuse, Ollama, Cohere, AWS, GitHub Actions, Tesseract, Pytest*

- Architected a **LangGraph-orchestrated multi-agent pipeline** of 9 specialized agents executing in parallel (6 concurrent), reducing **P90 latency by ~70%** to **sub-3-minute** response times under load.
- Engineered **LangChain** based agents with automatic **Ollama** to **Cohere** failover and retries, achieving **82.5%** extraction quality via **rule-based** validation combined with an **LLM judge**.
- Built **FastAPI** backend that ingests PDF/DOCX/images/TXT, runs **Tesseract OCR**, provides multilingual translation, and enforces rate limits plus retry policies.
- Integrated end-to-end tracing with **Langfuse**, **CI** with **GitHub Actions**, **PyTest**, deployed a cost-efficient **AWS EC2** PoC to eliminate per-token fees.

AI-Powered Mental Health Analysis System

Oct 2022 – Nov 2025

Tech Stack: *Python, FastAPI, Docker, AWS, GitHub Actions, PostgreSQL, Airflow, Mlflow, Hugging Face, PyTorch, TensorFlow, scikit-learn*

- Built **FastAPI** microservice using **Docker** on **AWS (ECR, EC2, RDS)** with **GitHub Actions CI/CD** pipelines, reducing API response time from **0.8s to 0.5s by loading models once at startup and using async throughout the request pipeline for non-blocking database operations and concurrent request handling.**
 - Orchestrated 3 ML training pipelines with Apache Airflow, implementing data versioning and model registration using S3, MLflow, and WandB.
 - Developed sentiment classification models using **scikit-learn** and improved depression detection performance (**TPR**) by **31.9%** through **BERT-based LoRA fine-tuning** via **Hugging Face** and **PyTorch** for early detection of cases.
 - Trained **CNN model for emotion detection on 20K+ augmented images using TensorFlow**, achieving **87% accuracy with real-time inference in <2s per image.**
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Skills

- **Programming & Tools:** Python, SQL, R, Git/GitHub, Linux
 - **ML/DL Frameworks:** PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, OpenCV
 - **Machine Learning:** Regression, Classification, Time Series, Clustering, Ensemble Methods, A/B Testing
 - **Deep Learning:** CNNs, Transfer Learning, Fine-tuning, YOLO, UNet, OCR
 - **NLP & GenAI:** LLMs, RAG, AI Agents, Transformers, BERT, LoRA/PEFT, Hugging Face, LangChain, Embeddings, CrewAI, LangGraph
 - **MLOps & Cloud:** Docker, Kubernetes, FastAPI, Flask, MLflow, CI/CD, AWS, GitHub Actions, Prometheus, Grafana, Langfuse, Loki, PyTest, Locust, Experiment Tracking, Prompt/Model/ Data versioning
 - **Databases:** PostgreSQL, MongoDB, Qdrant, Milvus, BigQuery, Redis
 - **Data Engineering:** Airflow, Dagster, Spark, Kafka, dbt, ETL/ELT, Web Scraping
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Mentorship & Community Leadership

HumaVolve

AI Mentor & Technical Instructor

- Led an intensive AI and Data Science bootcamp for fresh graduates and university students, teaching production ML lifecycles, RAG systems, and Docker/FastAPI containerized deployments.
- Mentored aspiring AI engineers through hands-on code reviews, architecture guidance, and pipeline debugging to ensure robust end-to-end project delivery.

Open Educational Resources & Community Contribution

Independent Initiatives | GitHub & Kaggle

- Authored the open-source "**AI and Data Science Lab**" repository, implementing foundational machine learning and deep learning algorithms from scratch in **Python** and **PyTorch** as reference implementations.
 - Published educational **Kaggle** notebooks and datasets, demonstrating advanced feature engineering, exploratory data analysis, and model optimization to help community practitioners master core ML concepts.
 - Cut annotation time by **90%** using **point-supervised semantic segmentation** on aerial images using **UNet** and a partial focal loss function.
 - Segmented 100M+ customers with **Spark K-Means**, revealing 97% checkout abandonment and reducing training from 40 to 15 minutes with PCA.
 - Applied **LightGBM** for time series demand forecasting across 75+ zones (4.5M+ monthly records, 20+ features), achieving a 60% MAE reduction.
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Education

Alexandria University, Egypt

Bachelor of Data Science and Computing

Sep 2019 – Jun 2023